

NITHISH BHARATHWAJ N

 03 March 2008  nithishbharathwajn@gmail.com  9363958388  Nithish Bharathwaj N  Nithish-Bharathwaj-N

Education

Bachelor of Engineering, *Chennai Institute of Technology* 09/2025 – Present
Computer science and Engineering (Cyber Security)

Summary

Full-Stack Developer & Cybersecurity Student with expertise in building AI applications and 3D digital twins using **React 19, TypeScript, Three.js**, and **Python (FastAPI)**. **Top 8 Finalist in Aerothon 2026 (HAL & IIT Indore)**, finalist in 5+ national hackathons, and **500+ LeetCode solver (Rating: 1771)**.

Skills

Programming Languages

- Python • C++ • C • JavaScript • TypeScript • SQL • Java

Frontend & 3D Visualization

- React 19 • Next.js • Three.js (WebGL) • Blender • HTML5 • CSS3 •

Development Tools & Cloud

- Git • GitHub • Docker • Linux • CI/CD • AWS Cloud (AI/ML) • Vercel • Vite • Postman

Cybersecurity

- Networking • Linux Administration • CTF (Capture The Flag) • Threat Modeling

Backend & Database

- Node.js • Express.js • FastAPI • RESTful APIs • Socket.IO  • MySQL • PostgreSQL

Core Concepts

- Data Structures & Algorithms • Object-Oriented Programming

Machine Learning & AI

- Scikit-Learn • Multi-Target Regression (Random Forest, Ensemble) • Foundation Models • LLMs • Prompt Engineering

Internship

Generative AI Virtual Intern, *AWS Academy × AICTE × EduSkills* 10/2025 – 12/2025

- Completed a **10-week virtual internship** on **Generative AI** and **AWS Cloud**.
- Worked with **LLMs, Prompt Engineering, Foundation Models**, and **AWS AI services**.
- Built foundational knowledge for developing **AI-powered cloud applications**.

Projects

SubAERO – Aerospace Digital Twin & PHM Platform 06/2026 – 08/2026

Tech Stack: React 19, TypeScript, Three.js (WebGL), Blender, Python, FastAPI, Scikit-Learn, Tailwind CSS, Vercel

- Architected an Aerospace Digital Twin & PHM platform** for HAL Tejas turbojet engines using **React 19, Three.js**, and **FastAPI**.
- Modeled 3D engine components in Blender** with real-time **WebGL visualization**, hotspot damage tracking, and aero-thermal physics calculations.
- Trained Multi-Target ML Ensemble models** in Scikit-Learn (**98.7%–99.9% R² accuracy**) predicting component health, thrust, and Remaining Useful Life (RUL).

Voyage AI – AI-Powered Intelligent Travel Planning Platform 07/2026 – Present

Tech Stack: React.js, Node.js, Express.js, Tailwind CSS, REST APIs

- Developed a full-stack **AI-powered travel planning platform** to generate personalized itineraries and simplify end-to-end trip planning.
- Implemented **AI-based itinerary generation**, destination recommendations, secure user authentication, and interactive trip management.
- Built a **responsive web application** with personalized dashboards and itinerary management for an enhanced user experience.

Certificates

- AWS Academy Gen-AI Virtual Internship (AICTE × EduSkills)
 - Cisco Operating Systems
- Cisco Python Essentials 1
 - Cisco Introduction to Cybersecurity
- Cisco Python Essentials 2
 - Cisco Introduction to Modern AI

Achievements

- Top 8 Finalist – Aerothon 2026 (HAL & IIT Indore)** • Finalist in **5+ National Hackathons** for AI & full-stack software solutions.
- 500+ LeetCode Solved (Rating: 1771)** • Maintained a **118-day coding streak** • International Abacus Competition qualifier.
- Sports & Leadership:** District-Level Throwball & Zonal-Level Carrom player • State-level athletics & cricket participant (CM Trophy).